ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 2

MECHATRONICS

PAPER 1 · PRACTICE A · MARKING GUIDELINE

MARKS: 100

Marking guideline for the Academic Central NC(V) Level 2 Mechatronics paper 1 · practice a.

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How to mark: each ✓ is one mark. Give marks for the formula, the substitution and the answer with the correct unit. Use consistent accuracy. Accept other correct answers and examples.

SECTION A

QUESTION 1

1.1 1.1.1 A ✓ 1.1.2 B ✓ 1.1.3 D ✓ 1.1.4 A ✓ 1.1.5 C ✓ 1.1.6 C ✓ 1.1.7 A ✓ 1.1.8 D ✓ 1.1.9 B ✓ 1.1.10 D ✓ (10)

1.2 1.2.1 True ✓ 1.2.2 False ✓ 1.2.3 True ✓ 1.2.4 False ✓ 1.2.5 True ✓ 1.2.6 True ✓ 1.2.7 True ✓ 1.2.8 False ✓ 1.2.9 True ✓ 1.2.10 True ✓ (10)

1.3 1.3.1 F ✓ 1.3.2 I ✓ 1.3.3 D ✓ 1.3.4 G ✓ 1.3.5 A ✓ 1.3.6 H ✓ 1.3.7 C ✓ 1.3.8 B ✓ 1.3.9 J ✓ 1.3.10 E ✓ (10)

TOTAL SECTION A: [30]

SECTION B

QUESTION 2

2.1 A system that combines mechanical parts, electronics (sensors and actuators) and a computer/controller to perform a task automatically ✓✓. Sliding door: input – a motion (PIR / microwave) sensor detects a person ✓; process – the controller decides to open the door and for how long ✓; output – a motor with a belt or rack opens and closes the door ✓; a limit switch / safety sensor gives feedback ✓ (6)

2.2 Any FOUR: wear safety glasses and safety shoes ✓; no loose clothing, jewellery or long loose hair ✓; switch off and isolate machines before maintenance ✓; keep the work area clean and tidy ✓; use guards on machines; know where the emergency stop and fire extinguisher are; use tools only for their purpose (4)

2.3 (a) tin snips ✓ (b) spanner / socket wrench ✓ (c) wire strippers ✓ (d) spirit level ✓ (4)

2.4 First class – fulcrum between effort and load ✓, e.g. seesaw, scissors, crowbar ✓; second class – load between fulcrum and effort ✓, e.g. wheelbarrow, bottle opener ✓; third class – effort between fulcrum and load ✓, e.g. tweezers, fishing rod, human forearm ✓ (6)

[20]

QUESTION 3

3.1.1 $\text{Load} \times 0{,}3 = 150 \times 1{,}2$ ✓; $\text{load} = 180 \div 0{,}3$ ✓ $= 600$ N ✓ (3)

3.1.2 $\text{MA} = \text{load} \div \text{effort} = 600 \div 150$ ✓ $= 4$ ✓ (2)

3.2.1 $\text{GR} = \text{driven} \div \text{driver} = 60 \div 20$ ✓ $= 3:1$ ✓ (2)

3.2.2 $\text{N} = 300 \div 3$ ✓ $= 100$ r/min ✓ (2)

3.2.3 The torque increases ✓ – about three times larger (ignoring losses), because speed is reduced ✓. (2)

3.2.4 An idler gear between them ✓ (1)

3.3 $\text{Effort} = \text{load} \div \text{number of supporting ropes} \checkmark = 800 \div 4 \checkmark = 200\ \text{N} \checkmark$ (3)

3.4 $\text{(a) W} = \text{F} \times \text{s} = 50 \times 1{,}5 \checkmark = 75\ \text{J} \checkmark\ \text{(b) P} = \text{W} \div \text{t} = 75 \div 3 \checkmark = 25\ \text{W} \checkmark$ (4)

3.5 Any TWO: lubrication (oil, grease) ✓; bearings ✓; smooth surfaces / polishing; correct alignment (2)

[21]

QUESTION 4

4.1 Battery symbol ✓; switch ✓; fuse ✓; motor (M in a circle) ✓; all in series in a closed circuit ✓ (5)

4.2 $\text{P} = \text{V} \times \text{I} = 24 \times 2 \checkmark = 48\ \text{W} \checkmark$ (2)

4.3 Any THREE: air is free and readily available ✓; clean – leaks do not cause oil spills ✓; fast movement ✓; cheaper components; safe in explosive or wet environments; no return lines needed (3)

4.4 When the button is pressed, the valve opens and compressed air flows into the cylinder ✓, pushing the piston out (outstroke) ✓; when the button is released, the valve exhausts the air to the atmosphere ✓ and the spring pushes the piston back (instroke) ✓. (4)

[14]

QUESTION 5

5.1 A sensor detects a physical quantity and converts it into an electrical signal (input) ✓; an actuator converts an electrical signal into movement or another physical action (output) ✓. Sensors (any TWO) ✓✓: LDR, thermistor, ultrasonic, limit switch, proximity sensor. Actuators (any TWO) ✓✓: motor, solenoid, servo, pneumatic cylinder, relay (6)

5.2 (a) inductive proximity sensor ✓ (b) LDR / light sensor ✓ (c) reed switch with magnet (or a micro switch) ✓ (3)

5.3 Input: light sensor (LDR) ✓ (or real-time clock); controller: microcontroller (e.g. Arduino) / PLC ✓; output: motor or linear actuator to open and close the door ✓ (limit switches may also be named) (3)

5.4 Moving parts wear and dust or dirt builds up, which can cause breakdowns and unsafe operation ✓. Any TWO tasks ✓✓: lubricate moving parts; check and tighten belts and chains; clean sensors; check wiring and connectors; drain water from the air compressor; replace worn parts (3)

[15]

TOTAL SECTION B: [70]

TOTAL: 100

END OF MARKING GUIDELINE